

A WISER intervention to combat the influence of misinformation on social media

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Abstract

Purpose – Misinformation is a major threat to individual, organizational and societal well-being. Combating it requires change on multiple levels. The purpose of this paper is to describe two co-created initiatives designed to holistically reduce the prevalence and effects of misinformation on social media.

Design/methodology/approach – The authors adopted a co-creation approach by convening a stakeholder group including social media, advertising, artificial intelligence and advocacy leaders. The authors also

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engaged social media users to collectively generate, test and implement an anti-misinformation strategy. These stakeholders shaped the development of the tools described in this paper.

Findings – First, the authors present a Social Media Ecosystem Map and Incentive Analysis that achieve impact by documenting avenues for action. Second, the authors illustrate impact at the user level by developing and disseminating the WISER framework, which delivers an implementable and memorable anti-misinformation strategy.

Research limitations/implications – Strategies tailored to the needs and characteristics of stakeholders are most likely to be persuasive. The discoveries and conclusions incorporate the contexts of the stakeholders with whom the authors collaborated. Future work will reveal recommendations for a broader audience.

Practical implications – The Social Media Ecosystem Map, Incentive Analysis and WISER framework represent promising foundations to spark novel insights and actions for ways to be WISER about information on social media.

Originality/value – Impact is often limited because strategies are developed in academic, corporate or policy silos. The authors adopt an original approach by exchanging knowledge among industry, academia and users with the aim of maximizing effectiveness and adoption.

Keywords Misinformation, Social media, Intervention, Wisdom, Online, Disinformation, Fake news, Transformative consumer research

Paper type Research paper

Though misinformation and disinformation have always existed, they spread further, faster and more easily today through social media [1]. Sixty-four percent of surveyed adults say that made-up news causes “a great deal of confusion” about current issues and events, and 23% admit to having shared false news themselves, often without realizing it (Barthel *et al.*, 2016). Beyond consumers, misinformation poses challenges for a variety of stakeholders, including social media companies that want to maintain public trust, brands whose reputations are put at risk when their advertisements are placed adjacent to dubious content on social media platforms and society when political candidates, their policies and democracy itself are called into question. These and other challenges will likely grow larger as advances in technology, such as AI, make misinformation more difficult to detect and deter. In response, researchers, policymakers and companies have proposed myriad interventions for addressing the influence of misinformation online. Yet, achieving positive impact is very difficult given the complexities of the problem and the multitude of stakeholders involved.

In this paper, we use two analysis lenses focusing on positive change in novel ways. First, we contend that an approach articulated by Keeling and Marshall (2022) illuminates new avenues for impactful action against misinformation on social media. Because the problem of misinformation is so vast and influenced by organizations and individuals with wide-ranging and competing incentives, *broad stakeholder engagement* is essential. These stakeholders and their viewpoints should inform a holistic understanding of the problem, and stakeholders should co-create solutions that are feasible, persuasive and effective.

Second, we argue that *wisdom* provides a unique and insightful lens to develop tools that can mitigate misinformation on social media (Schneider *et al.*, 2022; Ozanne *et al.*, 2021). Throughout history, wisdom has been seen as an essential concept and process for navigating life’s dilemmas and for fostering individual, organizational and societal well-being. Many definitions of wisdom exist; our application centers on Aristotle’s notion of “*phronesis*” or “practical wisdom,” which can be thought of as “the capacity to recognize the essentials of what we encounter and to respond well and fittingly to those circumstances” (Fowers, 2003, p. 415). Wisdom is distinct from constructs like intelligence because it incorporates elements of humility, emotion regulation, morality and prosociality. As such, it provides a particularly relevant lens for addressing misinformation on social media, which promotes echo chambers, strong emotions and – as former Google Ethicist and Center for Human Technology President

and co-CEO Tristan Harris warns – a “race to the bottom of the brain stem.” Consumers do not simply need to be smarter about misinformation; coping with the moral and well-being dilemmas brought forth by misinformation demands a need to be wiser.

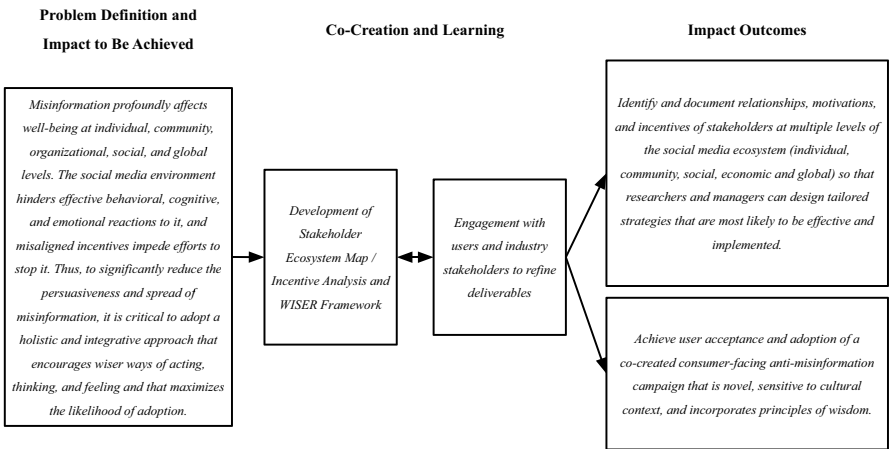
Thus, we pursued two objectives while defining and co-creating impact with stakeholders (see Figure 1). First, we engaged with industry experts to determine how influence (e.g. money) and information flow through the social media ecosystem and, therefore, what opportunities exist for partnerships between academic and nonacademic stakeholders to realize mutual impact. Second, we partnered with industry experts and social media users to co-create, apply, disseminate and test our WISER framework, which is an impactful user-facing intervention designed to:

- improve digital literacy by teaching users about platform technology;
- increase user self-awareness by highlighting how platforms capitalize on cognitive biases and habitual behaviors; and
- enhance consumer skills by teaching users how to protect themselves and others.

The intervention draws on wisdom theory and extant literature to provide a holistic, integrated framework and a unified strategy for combating the influence of misinformation on social media, as a complement to existing tools which are primarily cognitively focused (e.g. Caulfield and Wineburg, 2023). Hence, the novel contribution of the WISER Framework is to weave together multiple insights from various sources to develop a new holistic consumer-centric tool for addressing the dangerous contagion of misinformation through social media.

In the sections that follow, we will describe:

- how we worked with stakeholders to define the nature and scope of the problem and the impact to be achieved;
- what we learned by co-creating with nonacademic stakeholders;
- our two impact outcomes and the impact we achieved; and
- ethical considerations related to impact and driving change.



Source(s): Authors' own work

Figure 1. Overview of research

Working with stakeholders to define the problem and impact to be achieved

Identifying industry stakeholders and defining the problem was an iterative process. That is, uncovering various aspects of the problem with initial key players helped us to identify additional partners who could shed light on further elements of the problem. Given our focus on the spread of misinformation on social media, we started by identifying social media companies as key stakeholders. These early conversations highlighted the platforms' advertising revenue model as an important motivator of behavior, which, in turn, allowed us to identify advertising agencies as a key stakeholder group, and so on.

As we unfolded this process, we assembled an Expert Panel comprised of various industry stakeholders: a Quantitative UX Researcher and former member of a user safety team at a prominent social media platform; a former Senior Software Engineer who led the development of analytics tools at a major social media platform; a Director of AI ethics with experience at a major strategy consulting firm and a leading software firm; two advertising agency CEOs; a Senior Vice President in the advertising industry; and the Head of Identity at a software company.

We also met with individuals who represented key stakeholder groups in one-on-one meetings to supplement the insights that we gleaned from our Expert Panelists. These additional individuals included the Youth and Education Advisor at a nonprofit organization that seeks to align technology with human well-being, the Executive Director of a nonprofit organization that protects children from the influences of technology and advocates for them with policymakers and the co-Founder and co-CEO of a company that uses AI to measure adjacency of advertising and social media content.

Following our conversations with these stakeholders, we concluded that the spread of misinformation on social media is a complex, global issue that requires a multipronged, culturally-relevant approach at different levels. We also learned that stakeholder incentives are diverse and often misaligned, thus hampering change. Finally, we acknowledged that no single solution can tackle the totality of the problem. As such, we defined the scope of our investigation into the misinformation problem as follows:

Misinformation profoundly affects well-being at individual, community, organizational, social, and global levels. The social media environment hinders effective behavioral, cognitive, and emotional reactions to it, and misaligned incentives impede efforts to stop it. Thus, to significantly reduce the persuasiveness and spread of misinformation, it is critical to adopt a holistic and integrative approach that encourages wiser ways of acting, thinking, and feeling and that maximizes the likelihood of adoption.

Stakeholders were also critical for defining the positive impact to be achieved, as they helped us to develop a broader and deeper understanding of the social media ecosystem and identify potential leverage points for intervention. Ultimately, it became clear that all stakeholders have shared ownership over bringing about impact and that this shared ownership limits the likelihood of one group enacting change on its own. Co-creation between academic and nonacademic stakeholders is an important step for realizing positive impact at both the individual and systemic levels.

Co-creation and learning

Working with stakeholders

To facilitate co-creation between academic and nonacademic stakeholders, we held a series of meetings between members of the research team and the stakeholder groups. We sought to gather rich insights from a range of stakeholder perspectives and facilitate cross-stakeholder dialogue to generate collective knowledge. These meetings allowed all actors to toggle between macro and micro views of the problem, sometimes taking a high-level, integrative perspective involving all stakeholders and sometimes focusing on a particular stakeholder's

vantage point. Discussions ranged from individual conversations with the researchers to larger, more formal discussions involving a broader set of stakeholders. While the academic researchers facilitated the meetings and took detailed notes covering the insights gleaned, the nonacademic participants served the role of informants and collaborated on defining the problem, determining the impact to be achieved and developing potential solutions. Stakeholders also helped to identify other stakeholder groups as different aspects of the problem were uncovered. For example, an advertising executive helped connect us with a stakeholder at a content measurement platform. By the conclusion of these dialogues with stakeholders, two sets of knowledge emerged: procedural learnings that can guide future collaborative efforts (i.e. learning *about* co-creation) and problem-focused learnings concerning the influence of misinformation on social media (i.e. learning *from* co-creation).

Learning about co-creation with stakeholders

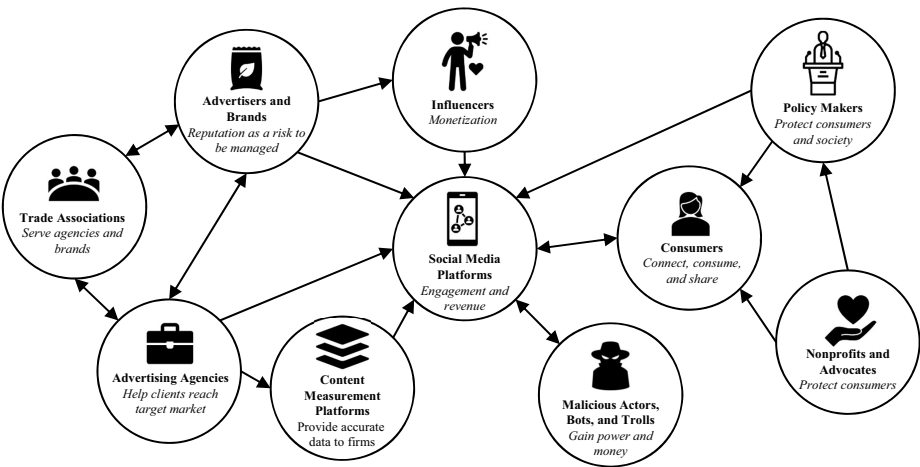
Our efforts to convene disparate stakeholders presented challenges while simultaneously illuminating procedural insights for future collaborations. The first challenge was representation. We sought to include members of an internal research team at a leading social media platform, but they ultimately decided that our work was not a priority. We mitigated this challenge by including two panelists who formerly worked in social media. The second challenge was integration. People working in different areas often develop their own vernacular and set of expectations about what is possible, inhibiting their ability to achieve a shared understanding. For example, timelines often differ; industry tends to move at a faster pace than academic, legal or political domains. We sought to establish common norms and practices so that all stakeholders could inform a cohesive plan for impact. The partners we engaged were critical sources of information about feasibility as well as the ethical and legal implications of possible courses of action. The third challenge was tensions between stakeholder groups. For example, some nonprofit and advocacy partners demonstrated skepticism toward advertising and tech industry stakeholders and were hesitant to be associated with industry money. We sought to promote a culture of transparency and a set of mutually beneficial outcomes. When reflecting on the co-creation process, one member of the Expert Panel remarked that, “The varied experiences and [points of view] created a conversation that clarified the most complex issues, and generated innovative ideas to address this crisis.” That is, broadly considering elements within the social media ecosystem highlighted interdependencies and differences in practices between actors, which helped to crystallize ideas.

Learning from co-creation with stakeholders

Critically, our conversations generated insights regarding how to influence the ways in which people cope with misinformation on social media. For example, the researchers learned a significant amount about the actors in the social media landscape and the relationships that drive their behavior, such as the importance of brand safety and how competing goals within an organization can hinder well-intentioned efforts to exact change. Conversely, the researchers shared perspectives on wisdom theory and the nature of human emotion, cognition and behavior. Together, these learnings yielded two Impact Outcomes: the Stakeholder Map/Incentive Analysis which takes a broad, holistic view of how stakeholders are interrelated and can address the problem and the WISER framework as one instantiation of that approach.

Impact outcomes: social media ecosystem map and incentive analysis

The initial outcome of our collaboration with the Expert Panel was a Social Media Ecosystem Map (Figure 2) and Incentive Analysis (Table 1). Formally, we defined Impact Outcome 1 as:



Note(s): Stakeholder motivations on social media are described in italics underneath each stakeholder. Arrows represent the flow of influence, money and information and signify levers for impact; for instance, through changing incentives, information flow or information processing

Source(s): Authors’ own work

Figure 2. Social media ecosystem map and avenues for impact

Identify and document relationships, motivations, and incentives of stakeholders at multiple levels of the social media ecosystem (individual, community, social, economic, and global) so that researchers and managers can design tailored strategies that are most likely to be effective and implemented.

The multifaceted nature of the social media industry means that there are numerous potential strategies to combat misinformation, each with a distinctive set of opportunities and limitations. Further, the likelihood that a particular tool will achieve a positive impact often depends on whether it is aligned with the incentives of numerous stakeholders. Thus, we first took a wide-ranging view of the industry and documented factors that drive the behavior of individual stakeholders.

The resulting Ecosystem Map illustrates key relationships and flows of (mis)information and influence (e.g. money, political pressure and user-attention) between stakeholders in the social media misinformation ecosystem. The Incentive Analysis details incentives that motivate behavior for each stakeholder in the network, “input levers” that could prompt that stakeholder to act, and “output levers” that represent tools available to that stakeholder to fight misinformation both directly and indirectly. In sum, the Ecosystem Map and Incentive Analysis provide guidance for stakeholders regarding which actors have power in this system, how they are related to each other, which incentives drive their behavior and the opportunities and limitations they face in relation to combating misinformation.

We believe that the Ecosystem Map and Incentive Analysis will serve as a valuable resource for organizations, researchers, policymakers and others to focus their anti-misinformation efforts in the most pragmatic way possible, taking into account the unique perspectives, resources and constraints associated with each stakeholder group.

Table 1. Stakeholder incentive analysis

Stakeholder	Motivations	Input levers <i>i.e. What levers are available for stakeholders to persuade them to reduce misinformation?</i>	Output levers <i>i.e. What levers are available for them to reduce misinformation?</i>
Malicious actors, bots, trolls	• Financial benefit • Exercise influence over relevant issues (e.g. political outcomes, product purchases, etc.)	• These actors purposefully spread false information; they are unlikely to stop willingly. They often conceal their identity, so moderating content/account removal is difficult. Legal actions may be possible but difficult to enforce as actors operate globally and speech is protected in many areas	• None due to incentive incompatibility
Social media platforms	• Profitability • Acquiring, maintaining and profiling an active user base to generate engagement and ad revenue • Fulfilling more consumer needs (e.g. platform seen as a news source in addition to source of social connection) to more deeply engage users	• Users can disengage from the platform (e.g. canceling their account, boycotting). • Brands can shift ad spending to other avenues in response to brand safety threats. However, coordinated action is not permitted • Governments can regulate platforms (e.g. by mandating action on content moderation). However, regulation may lead to harmful censorship, and speech is protected in many areas	• Preexposure warning/content inoculation campaigns (e.g. YouTube's "hit pause") deliver general information about misinformation and teach anti-misinformation strategies. However, users may not trust platforms as credible messengers of such guidance, feel that it is patronizing, or forget the warnings/actions • Specific content that violates policies (e.g. incites violence, hate speech, etc.) can be removed or deprioritized in recommender algorithms. However, detection of such content can be difficult, and platforms may be subject to accusations of censorship • Some platforms (e.g. Instagram) display warning labels when users are confronted with content that is potentially false (e.g. telling users that the content has been deemed false by fact checkers or sharing additional contextual information) or made/modified by AI. Content could be flagged

(continued)

Table 1. Continued

Stakeholder	Motivations	Input levers i.e. What levers are available for stakeholders to persuade them to reduce misinformation?	Output levers i.e. What levers are available for them to reduce misinformation?
Influencers	• Gain large and engaged user following • Financial benefit • Influence/user trust • Helping users	• Brands seek reliable partners who fit their image/target audience. Brands can evaluate the veracity of influencer content, establish conduct codes and end partnerships with influencers who spread misinformation • Other influencers can confront content creators who propagate misinformation and promote those who create true content	for labeling by human moderators or AI. However, users may experience reactance and be more likely to share information they feel is being unfairly repressed. Platforms may also have censorship concerns • Platforms can provide positive labels to signal trusted information sources (e.g. YouTube tells users that videos posted by some universities come from an accredited healthcare educator) • Users can be removed/blocked/restricted, though platforms may avoid doing so because they are motivated to keep a large user base • User controls (e.g. blocking/filtering) may be effective, but users must notice and act upon them. Controls can reduce engagement and platforms' control of content, so platforms may be hesitant to implement them
			• Influencers should ensure that they cultivate engagement using content that is truthful and not overly sensationalized • Influencers can use their visibility as well as their familiar and trusted status with users to provide education and encourage adoption of good digital media literacy habits. Influencers who consider such messages to (continued)

Table 1. Continued

Stakeholder	Motivations	Input levers <i>i.e. What levers are available for stakeholders to persuade them to reduce misinformation?</i>	Output levers <i>i.e. What levers are available for them to reduce misinformation?</i>
Advertisers/brands	- Achieving marketing objectives (e.g. awareness, liking, sales conversions or referrals) - Educating consumers (e.g. government and nonprofit groups) - Brand safety (i.e. ensure that brand does not appear alongside objectionable content)	- Users can subscribe and interact with influencers who promote truthful content and disengage from those who do not - Trade groups (e.g. European influencer marketing alliance, American influencer council) can educate/enforce ethics codes - Customers can express disapproval by shifting purchases to competitors or public comment (e.g. posting on social media) - Ad placements occur via algorithms controlled by social media platforms. Content measurement platforms can help brands detect and avoid problematic placements - Trade organizations can educate brands about misinformation, platforms' performance at controlling it and ways to reach consumers when platforms do not maintain brand safety, though coordinated activity is not permitted in some jurisdictions (e.g. USA)	- be authentic to their "brand" may be ideal partners - Organizations can shift spending away from platforms or content creators behaving in ways that might harm the brand. However, such behavior has historically been ineffective at changing platform behavior or has spurred accusations of collusion - Brands can create anti-misinformation messaging, especially those brands that are oriented toward trust and veracity. However, efficacy may be compromised by the activation of persuasion knowledge in users - Brands can support nonprofits and research via financial/intellectual resources. However, such support may compromise their independence
	- Ad agencies: Acquiring/servicing/retaining clients by facilitating marketing objectives (e.g. sales, etc.) - Trade associations: Advancing the goals of membership constituencies (ad	The route to persuade action by advertising agencies is largely through clients, as they are ultimately the key decision-makers. Clients can inform agencies that they are concerned about the impacts of misinformation and want to take concrete steps to combat it	These groups can serve a critical advisory role in providing brands with informed counsel from a broader perspective. In particular, setting common, industry-wide definitions and standards for misinformation, brand safety and mitigation strategies may help advertisers, social media

(continued)

Table 1. Continued

Stakeholder	Motivations	Input levers i.e. What levers are available for stakeholders to persuade them to reduce misinformation?	Output levers i.e. What levers are available for them to reduce misinformation?
Policymakers (politicians and government agencies such as the European data protection supervisor, US federal trade commission, etc.)	agencies, social media platforms, influencers, etc.)	Trade associations can provide industry-wide perspectives to agencies and clients, but they do not make advertising decisions. Thus, advertisers and agencies are the most effective parties to influence them	platforms and advertising agencies to realize more predictability and effective marketing outcomes
	Measurement platforms: Providing advertisers with accurate data on content adjacent to ad placements	Measurement platforms can offer brand safety tools in response to client demand	
	Politicians: Prioritizing/advancing policy aims and constituent interests; building constituent/donor support	Consumers can voice their concerns about misinformation to policymakers, though many competing initiatives vie for policymakers' attention	- Governments' capacity to act is often limited - Legislative/policy processes move more slowly than technology. For example, US law establishing liability protections for platforms that host objectionable content predates the existence of social media. Dozens of reforms establishing liability for platforms hosting and restricting content have been proposed since 2019, but many have failed to pass
	Agencies: Balancing needs stakeholders, protecting vulnerable parties, providing consistent policy, etc., within the policy mandate of the government administration	Organizations such as trade associations, nonprofits and advocacy groups can exert lobbying efforts on policymakers to take action against misinformation	- Mandating action (e.g. directing platforms to moderate content) can be seen as censorship - Often, speech is protected by law - Some government agencies set out expectations and connect industry stakeholders to develop common standards and self-enforced commitments (e.g. the

(continued)

Table 1. Continued

Stakeholder	Motivations	Input levers i.e. What levers are available for stakeholders to persuade them to reduce misinformation?	Output levers i.e. What levers are available for them to reduce misinformation?
Nonprofits/advocacy groups			European Union code of practice on disinformation and the Australian code of practice on disinformation and misinformation)
	Protecting consumers from the harmful effects of misinformation	- Nonprofit groups are already motivated, but they often lack resources or expertise - Individuals or organizations who have been harmed by misinformation can share stories with advocacy groups to vividly illustrate the problem that misinformation poses - Researchers can partner with nonprofits to share knowledge about the psychological and economic realities of the social media infrastructure and to disseminate messages to stakeholders like policymakers - Advertisers/platforms/nonprofits can collaborate to create anti-misinformation campaigns, though some actors may avoid relationships to maintain independence	- Policymakers can enact curriculum reforms, requiring digital media literacy education - Nonprofit organizations can educate consumers and share strategies to effectively behave online because they are seen as relatively unbiased - Nonprofit organizations can place and maintain pressure on policymakers, advertisers and platforms
Consumers	- Some motivations of consuming and sharing social media content include: - Amusement - Social interaction	Nonprofit organizations and advocacy groups can educate consumers on digital media literacy; however, user behavior on social media is often habitual and driven by heuristics, making the detection of misinformation difficult	- Users can use passive strategies (e.g. ignore content) or active strategies (e.g. investigate additional sources) - Individual users are relatively low-power members of the ecosystem, which is largely set up to influence them. They are the only stakeholder in this map for whom social

(continued)

Table 1. Continued

Stakeholder	Motivations	Input levers i.e. What levers are available for stakeholders to persuade them to reduce misinformation?	Output levers i.e. What levers are available for them to reduce misinformation?
	- Demonstration of group membership - Self-presentation (e.g. feeling smart) - Gathering information to facilitate decision-making - Altruism (this emerged as a frequent theme in our exploratory data; users often interact with and share content because they want to help inform others)	• Social media companies want to create a seamless user experience and are unlikely to implement anti-misinformation tools that interfere with the benefits of social media. Offline tools likely have more potential. The tools must be understandable, memorable and usable, and avoid being intrusive or patronizing • Interventions targeted at consumers must address the fact that platforms are designed to maintain engagement, minimize thought and maximize emotional resonance	media activities are not their job. Tools designed to help users take action against misinformation must be empowering, inform users about the motivations of other stakeholders and provide them with actionable guidance

Source(s): Authors' own work

For example, individual-level efforts have the advantage of directly influencing the most susceptible group of stakeholders—users. Equipping people with tools that help them to make choices that further their well-being while also maintaining free access to information is beneficial to all parties. However, individual users have a very diverse set of needs, information processing capacities and motivations. Reliably capturing their attention and advancing behavior is a significant challenge. Organizational-level levers – such as those available to social media platforms – can be effective, as platforms directly control the availability and reach of misinformation. However, social media organizations are typically focused on maintaining user engagement and profitability; tools that might disrupt those objectives are unlikely to be implemented in impactful ways. Finally, while system-level strategies such as shifting spending away from problematic platforms may seem promising – after all, financial incentives are powerful – there are many challenges with implementation. It is difficult for individual advertisers to have impact, coordinated action among them is not possible and such steps must ultimately align with profit so that firms can fulfil their fiduciary duty. Meanwhile, policy solutions have both wide-ranging impact and long-term potential but are slow and potentially hamstrung by concerns about censorship and constitutionality. Thus, each of these levels of action presents different opportunities and challenges for different stakeholders.

As academic researchers with expertise in developing an understanding of human affect, cognition and behavior and with a more limited capacity to drive organizational-level or system-level change, our conclusion from the Stakeholder Map and Incentive Analysis was that users should be our primary focus. As such, we defined Impact Outcome 2, highlighting the unique value that marketing academics with theoretical and pedagogical expertise provide when co-creating solutions in partnership with other stakeholder groups:

Achieve user acceptance and adoption of a co-created consumer-facing anti-misinformation campaign that is novel, sensitive to cultural context, and incorporates principles of wisdom.

Our discussions on formulating a user-facing intervention began with the creation and analysis of two prototype memes which we anticipated might help interrupt consumers' in-the-moment, automatic information processing and habitual sharing of information on social media. However, the Expert Panel was immediately skeptical of this strategy. They questioned whether users would be able to overcome impulsive tendencies when confronted with misinformation on social media. Furthermore, they noted that social media algorithms would likely deprioritize content aimed at creating cognitive friction, thus, hindering organic spread of the memes.

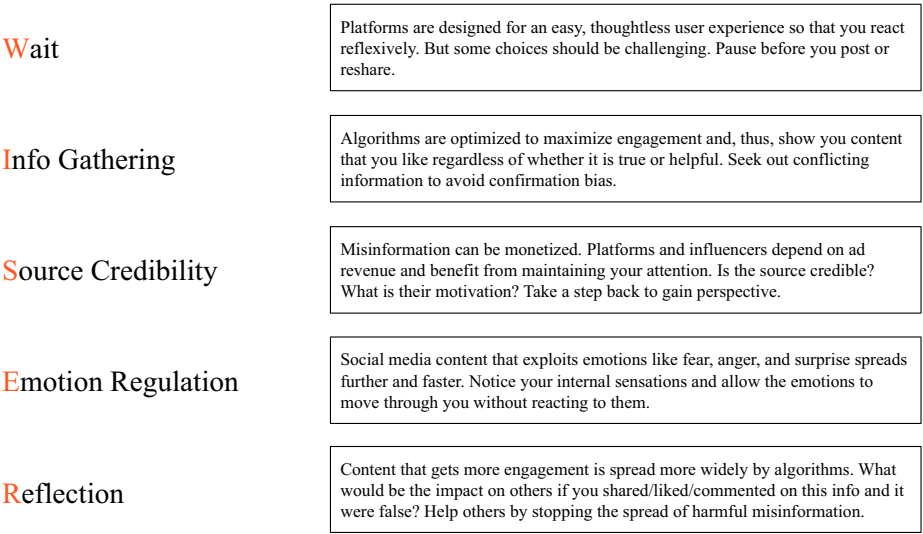
Instead, members of our panel suggested that an educational campaign designed to equip consumers with effective strategies against misinformation might be more successful. Six key success factors emerged from our discussions with experts. First, to enhance attention and persuasion, users should process educational information while they are in a “cool” state, not when they are confronted with arousing content. Second, interventions should incorporate both a behavior interruption strategy and an action plan. Because habit is a major driver of misinformation consumption (Ceylan *et al.*, 2023), it is important to prompt disengagement from maladaptive automatic responses in favor of preplanned response strategies that can easily be implemented (Gollwitzer, 1999). Third, the action plan should incorporate behavioral, (meta)cognitive and affective elements. Our natural responses to misinformation are typically rooted in strong emotions such as fear, disgust and surprise (Vosoughi *et al.*, 2018). Productively channeling those feelings is critical to motivating desired behaviors. Fourth, behaviors are more likely to be adopted if consumers understand why they are necessary. Thus, it is important to educate users about the specific mechanisms

that platforms use to generate engagement (e.g. algorithmic personalization). Fifth, many users do not engage with content on social media by sharing, liking or commenting. Yet, misinformation still guides their judgments and feelings. Thus, an intervention should cover a range of possible outcomes, including belief formation, which can influence offline consumer behaviors and information transmission. Sixth, it is critical to consider a mix of platform tools, paid media and organic spread as means for disseminating the intervention.

From these crucial lessons, our WISER framework emerged (see [Figure 3](#)). The WISER framework integrates (a) knowledge from our Expert Panel, particularly regarding how and why (mis)information spreads on social media (e.g. platforms are designed for seamless experiences, resulting in reflexive action) with (b) theoretical considerations, especially in relation to wisdom. As such, this integrative effort addresses the ways in which social media’s technological design capitalizes on limitations of human cognition, resulting in the proliferation of misinformation.

Impact outcomes: WISER framework

The WISER framework attempts to convey a holistic strategy for users to cope with misinformation. It directs users to *Wait*, engage in *Information gathering*, consider *Source credibility*, practice *Emotion regulation* and *Reflect* on the potential harm caused by their actions. The framework features a mix of behavior interruption, action plan and platform literacy content. Importantly, it is designed to be encoded in a “cool” state but remain highly memorable when consumers are faced with arousing content. This delivery enhances opportunities for reflection and developing emotion regulation strategies, which are important components of wisdom theory ([Grossmann et al., 2020](#)). Finally, our earlier work suggested that consumers share misinformation for prosocial reasons, such as the belief that



Source(s): Authors’ own work

Figure 3. WISER framework

it will help family, friends and society. Thus, the WISER framework also evokes prosocial values by asking consumers to consider the impact on others if the information they share turns out to be false.

After our knowledge exchange with the expert panelists, we summarized our discoveries related to the framework, integrated the practitioners' knowledge with academic research on wisdom and sought feedback from the experts to ensure that we had accurately captured the full range of insights. Based on their suggestions (e.g. avoid paternalistic language), we revised the framework before testing its impact.

Curriculum development and initial testing

We developed a 45-min live, interactive curriculum to introduce the WISER framework to social media users. We deployed a pilot version of this curriculum with eight social media users at a large US university. University is a time of transition, providing avenues for breaking existing habits and establishing new ones. The curriculum asked participants to reflect on how misinformation shows up in their lives and how they could use the WISER framework to make better decisions regarding what content to believe and share with others. During our conversation, users evaluated the framework positively, commenting on its ease of use, relevance and potential to propel behavioral change.

One week later, seven users accepted our invitation to complete a follow-up questionnaire regarding what they learned in the session and their social media usage during the intervening week. Six users recalled – without aided memory – between two and five of the five WISER dimensions. Perhaps more importantly, six users indicated that learning about the framework changed the way in which they engaged with social media content. Each element of the framework was mentioned as exerting an impact upon their thoughts, feelings or behaviors over the course of the week. For instance, one user wrote, “I believe I am a lot more cautious and careful when seeing a crazy post. I check the comments, I check who posted the article, I am just a lot more secure.” Consistent with the emotion regulation dimension of the framework, another remarked, “I have probably engaged in more emotional regulation, especially because viewing other people’s misinformation can be discouraging and frustrating.” Echoing the source credibility dimension, one user shared, “I have checked the credibility of sources a lot more than usual.” Finally, in line with the reflection component, another commented, “I began to question my true impact and even found myself unliking a video because I realized others would see it on their feed and could misinterpret it.” These discussions provided preliminary evidence that the WISER framework is memorable and can effectively drive improved behaviors surrounding social media misinformation.

Social media literacy event dissemination

After conducting the initial pilot of our educational curriculum and establishing its ability to generate impact, we sought to scale our impact by launching the WISER framework with approximately 500 high school students (aged 14–18), parents and educators as part of a larger event on social media literacy and well-being hosted with Social Awakening, an advocacy organization. The event featured two keynote presentations by the organization’s founder, a former media strategist who now educates youth about avoiding the negative effects of social media use.

Co-creating collateral for the event. We engaged a group of pending audience members in creating materials to deliver the WISER framework at the social media literacy event. First, we worked with five students from the focal high school to design a postcard that would resonate with this audience [2] (see Figure 4). This card serves as a visual anchor for



Source(s): Design by Keelyn Arnold

Figure 4. WISER framework postcard

the framework and can be used as an ongoing reminder. Students discussed the postcard's wording, design and cultural inclusivity, and we hired a professional graphic designer to create the postcard based on the students' input. A Spanish version was translated and back translated by native Spanish speakers.

Second, in collaboration with the students, we created a website that expands on the elements of the WISER framework and provides users with related resources, including a video co-created with social media users based on guidance from the Expert Panel.

Distribution of framework at the event. As attendees entered the event auditorium, they found the postcard on their chair summarizing the WISER framework in English on one side and Spanish on the other, along with a QR code linking to the website. The author who organized the event observed that most students and parents read the postcard, held it throughout the event and took it with them when they left. A school administrator also distributed postcards to the students who could not attend the presentation. The keynote speaker kept a postcard and offered to share the information within his network.

Curricular iterations and larger-scale testing

Subsequently, we sought to develop and test shorter, more scalable versions of the live presentation. We first consulted with the teaching and learning center at a US university for guidance on refining the presentation and enhancing the framework's scalability. We also returned to the high school where we held the social media literacy event to co-create a 10-minute version of the presentation with a small group of representative students and a high school staff member. We chose 10 minutes based on the staff member's guidance and constraints in the high school's schedule.

We delivered the shortened presentation to 185 high school students across eight sessions in two days (see Web appendix A for additional methodological details). The following week, those students who had submitted a signed parental consent form ($n = 32$) received a follow-up survey via email. Fifteen students participated. Nearly 30% remembered the name of the framework. Between 40% and 80% of students reported that they engaged in one or more of the WISER dimensions “a little more” or “a lot more” since the presentation, and 67% reported that learning about the framework was empowering (see Table 2).

In implementing this presentation, we observed the challenges of balancing learning objectives with feasibility. While the 10-minute presentation format was necessary to accommodate the time available in the high school’s schedule, we felt rushed delivering the presentation and did not have time to facilitate interactive, in-depth learning about each dimension of the WISER framework, per the teaching and learning center’s guidance.

Thus, we subsequently designed a 20-minute iteration of the curriculum to allow more time to engage users with the content. We deployed and tested this iteration with 103 social media users across seven sessions in two days at a large US university. Over two-thirds of users were able to remember the name of the framework three to four days later. In addition, 50–75% reported engaging in one or more of the WISER behaviors “a little more” or “a lot more” since the presentation, and about three-quarters said that they found learning about the WISER framework to be empowering (see Table 2).

From this experience, we concluded that iteration is often necessary to accommodate unique features of the deployment context. While preliminary evidence suggests that our framework can be effectively deployed within 20 min, creative solutions may be necessary to capitalize on shorter occasions to interact with users. We also acknowledge that, while these initial results are promising, a larger and more systematic sample will be needed to reliably test outcomes. Our future work will examine modularizing the WISER framework to present selected elements in shorter “chunks” to a broader audience.

Routes for future impact

Collaborating with a diverse range of stakeholders generated positive impacts with students, parents and school administrators, and it cultivated new mutually-positive relationships within an engaged network of industry leaders. Some of our stakeholders have continued to connect us with other potential collaborators, paving the way for a continuation of this work beyond the current project. We believe that four avenues hold particular promise for both applied impact and future research. First, countering misinformation requires comprehensive action by many stakeholders using varied tools. Our initial efforts focused on the WISER framework as a consumer-facing intervention with specific affective, cognitive and behavioral objectives. The Stakeholder Map and Incentive Analysis highlight many more ways for other organizations and individuals to tackle misinformation from different angles. As academic researchers, we found value in the process of engaging with industry experts who have diverse perspectives. We hope that the Incentive Analysis inspires other cross-functional collaborations between stakeholders with complementary skills and influence. Second, additional iteration and testing of the WISER framework is needed to further validate and extend our initial results. We aim to test the framework in multiple ways. A lab experiment offers the ability to evaluate the individual components in a controlled setting. In addition, we are currently discussing WISER framework deployment opportunities with a US state’s Attorney General’s Office and a state Senator who co-sponsored a bill to advance anti-misinformation/disinformation curricula in public schools. We are also in conversation with a national youth advocacy foundation that has expressed interest in collaboratively modularizing and scaling our WISER framework curriculum as part of online pedagogical

Table 2. User engagement with WISER framework following larger-scale testing at high school and large university

	Percentage of students reporting that they engaged in WISER behaviors “a lot more” or “a little more”			
	Since learning about the framework		University sample (n = 103)	
	High school sample (n = 15)	University sample (n = 103)	Representative quote	
	%	%		
Wait	60	68	“I waited and took some time to think before commenting or having opinions”	“[...] I saw a post about a major policy change recently, and instead of freaking out or sharing it right away, I decided to wait and not jump to conclusions”
Information gathering	53	51	“The other day, I was on social media when a video popped up talking about the wedding of two artists; I paused the video, searched on google about the wedding, and looked at many websites that confirmed that the wedding was fake news, but they were a current couple”	“[...] this weekend there was news about [former U.S. President Donald Trump's] second attempt assassination [...] [...] I looked and gathered for information or facts, checked if it was true...”
Source credibility	80	65	“[I] made sure it wasn't AI”	“I made sure that there were many credible sources all saying the same thing [about the attempted assassination of former U.S. President, Donald Trump]”
Emotion regulation	40	61	“When I saw something that might upset me, I put my phone down and did something else”	“I saw rage bait on [T]witter and chose not to engage with it and instead turn off my phone and study for my midterm. I chose to regulate my emotions by focusing on something actually productive and I thought about how embarrassing a response to the material would have been, given that it was made to evoke a reaction”
Reflection	53	70	“When seeing something that I knew was false but was believable enough for others to trust it, I didn't engage so my friends wouldn't”	“Every time that I am going to post something, I always try to read everything I wrote, and think about how it could affect others”
Percentage of students who remembered the WISER framework (unaided recall of the framework's name)	27	69		
Percentage of students who reported that learning about WISER made them feel “empowered”	67	74		
Source(s): Authors' own work				

materials available to teachers across the USA. We aim to launch a large-scale field study to further validate the intervention and measure its effectiveness longitudinally. This work would also allow us to test the framework across different formats (face-to-face delivery vs web-based interactive modules) and ensure that the framework does not spur unintended consequences, such as reactance. Third, we see potential in deploying the WISER framework directly to users outside of the educational context via advertising or social media. For example, components of the framework could be presented next to news content or incorporated into interactive games, allowing for a test of whether it enables users to better detect false news and for an examination of other moderating variables, such as age cohorts. Finally, while we have focused our initial efforts on combating misinformation, sometimes consumers create or disseminate information that they know to be false (i.e. disinformation). We argue that the WISER framework may also be useful in combating disinformation when users have good intentions – for example, when they knowingly share false content in an attempt at humor. When people are well-meaning, elements of the framework like behavioral interruption, emotion regulation and reflection should facilitate positive behaviors. It is valuable to test how the framework guides evaluations and behaviors (e.g. persuasion, intentions to share) when people are unaware that information is false, when they suspect it might be false or when they know it is false.

Ethics of impact

It is critical to consider the current and long-term ethical issues regarding:

- which stakeholders were able to contribute to defining impact and the co-creation process;
- the fuller implications of change; and
- the implied commitments among stakeholders.

First, not all stakeholders equally contributed to the co-creation of our intervention. The most powerful stakeholder – the platforms – chose not to engage with us at this time. While we invited a major platform's responsible research team to collaborate and they agreed to evaluate whether doing so would be a priority, they ultimately ceased communication. Indeed, many of the stakeholders with whom we collaborated reported struggles working with the platforms and noted that different teams within the same company often have competing priorities, making coordinated buy-in difficult.

At the same time, there are also ethical and inclusivity implications of engaging with individuals at the other end of the power structure – users. Our tool asks social media users to assume additional cognitive, affective and behavioral load in a social media context that already demands a great deal of self-regulatory resources, thus, placing an additional burden on the populations our intervention aims to serve. However, an individual-level intervention also offers a way to build trust within consumers, which may be particularly important during a “post-truth” era when users are skeptical of institutions, social media platforms and other parties providing guidance (e.g. warning labels). Indeed, most consumers reported in our follow-up surveys that they found the intervention to be empowering, and they demonstrated success in being able to implement it in real time. Future research could explore whether this new sense of self-reliance and self-confidence leads to more sustained and resilient behavior over time.

Another ethical concern is that our work is limited by the specific consumer populations with whom we engaged. In this case, we co-created and disseminated our framework with a largely Hispanic, first-generation student population. Even though we translated the postcard

into Spanish, it was developed by English-speaking academics and stakeholders. While we believe that the core of our framework is relevant across diverse consumer groups, the mode of delivery, the look and feel of the presentation and the specific behaviors in which consumers can engage to practice each WISER dimension will need to be modified based on consumer sociocultures, psychographics and demographics. Finally, regarding the ethics of focusing on consumers, a member of our Expert Panel raised the critical point that there may be reactance among consumers if they feel that the framework is dictating what they *should* do. Similarly, consumers may become defensive if they feel that the framework casts them as part of the problem. When sharing the framework with consumers, it is important to frame the risks and effects of misinformation as something that impacts everyone, particularly because of the profit and power that are at stake.

A second area of ethical concern is how stakeholders work together. Collaborating with nonacademic stakeholders involves a number of implied commitments between the stakeholders both during and following the co-creation process. To affirm emic validity, we shared drafts of the manuscript with our Expert Panel to make sure that we accurately captured the co-creation process as they experienced it. Regarding long-term commitments, many solutions to the problem of misinformation were proposed throughout the project's duration. While we focused on one particular project here – a behavior change campaign – stakeholders have discussed a desire to pursue increasingly more comprehensive approaches to combating misinformation in the future.

Finally, there is the issue of intellectual property and who owns the co-created WISER framework. To grow its impact, we offer the prospect of partnering with additional stakeholders to package and disseminate the framework within their networks. Because the spread of misinformation on social media is a multifaceted problem requiring a multipronged solution, it is critical to have a diverse team trying to address this issue from different angles. Thus, we are more interested in reach than we are concerned about ownership.

Conclusion

With the input of a diverse set of stakeholders, including industry leaders, social media users and educational institutions, we have co-created a positive impact paradigm for mitigating the deleterious influences of social media misinformation. These efforts have already resulted in two outcomes: a map of avenues that nonacademic and academic stakeholders can take and the fuller construction of one of those avenues via the inaugural WISER framework. Numerous opportunities for additional positive impact remain to be identified and realized, and the need to do so is urgent. We hope that our work inspires new insights and actions on ways to be WISER about living with misinformation on social media.

Notes

1. While misinformation is false or misleading information, disinformation has the added distinction of being created or disseminated with an intent to deceive. This paper is primarily concerned with facilitative mechanisms for users to cope with false information regardless of whether it was initially created with the intent to deceive; thus, for simplicity, we generally use the term “misinformation” throughout the manuscript. We note that our work addresses responses to both misinformation and disinformation and discuss those applications on p. 15.
2. The high school is located in a major US city and predominantly serves first-generation students from Spanish-speaking families.

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Supplementary material

The supplementary material for this article can be found online.

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